



Convolutional Neural Networks

Image Classification

- **X, O Classification**



A two-dimensional
array of pixels

Image Classification

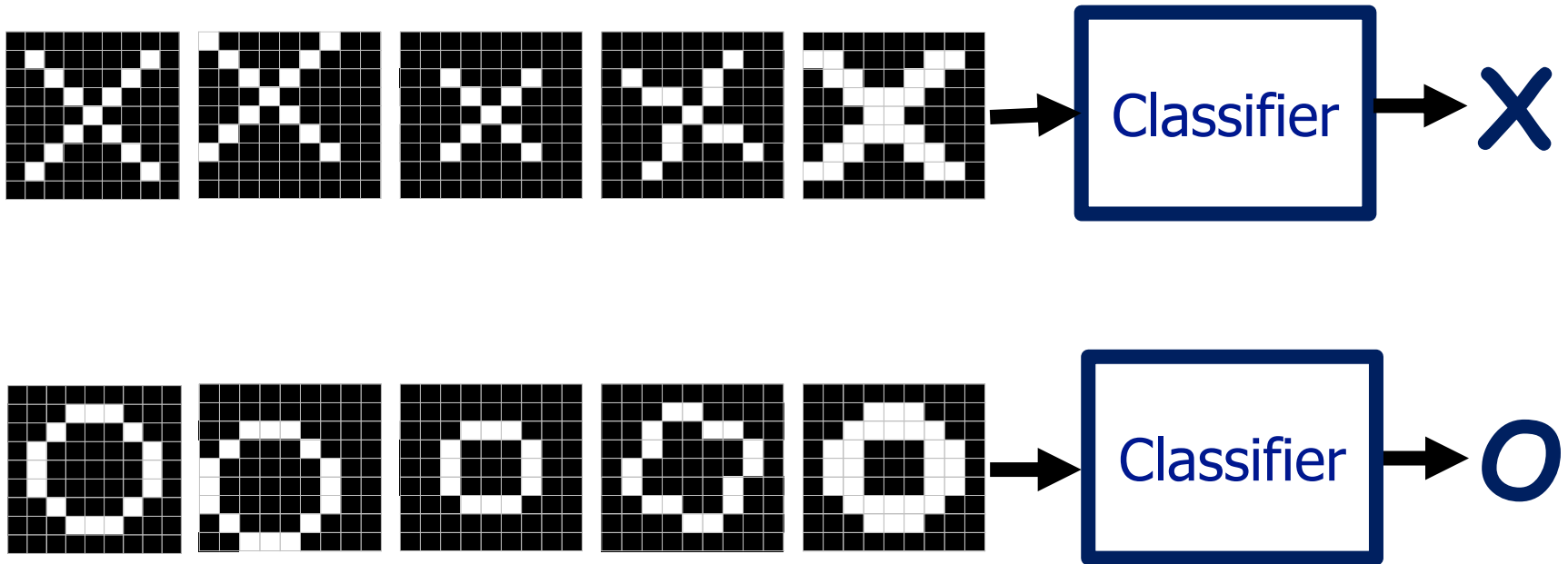
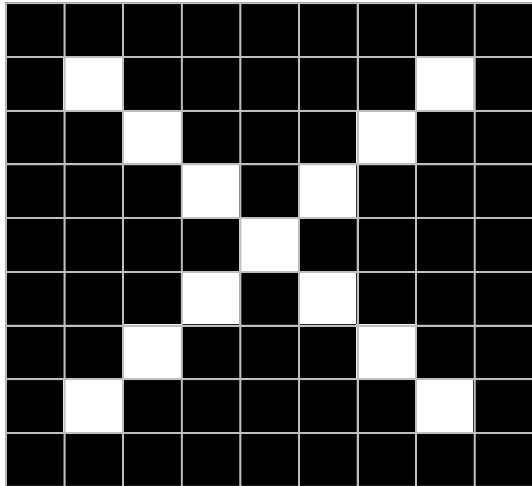


Image Classification

- **Same?**

- How to determine both are the same?



=

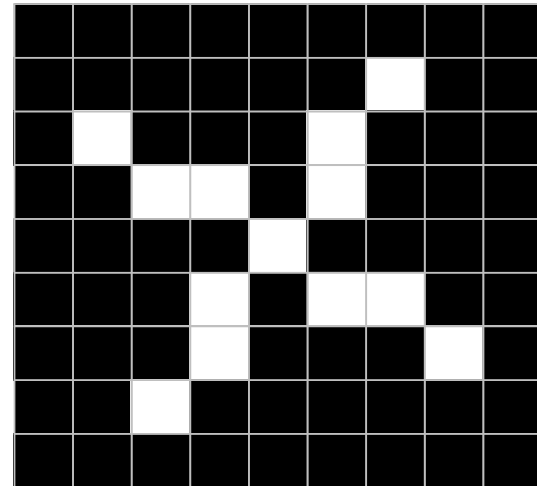
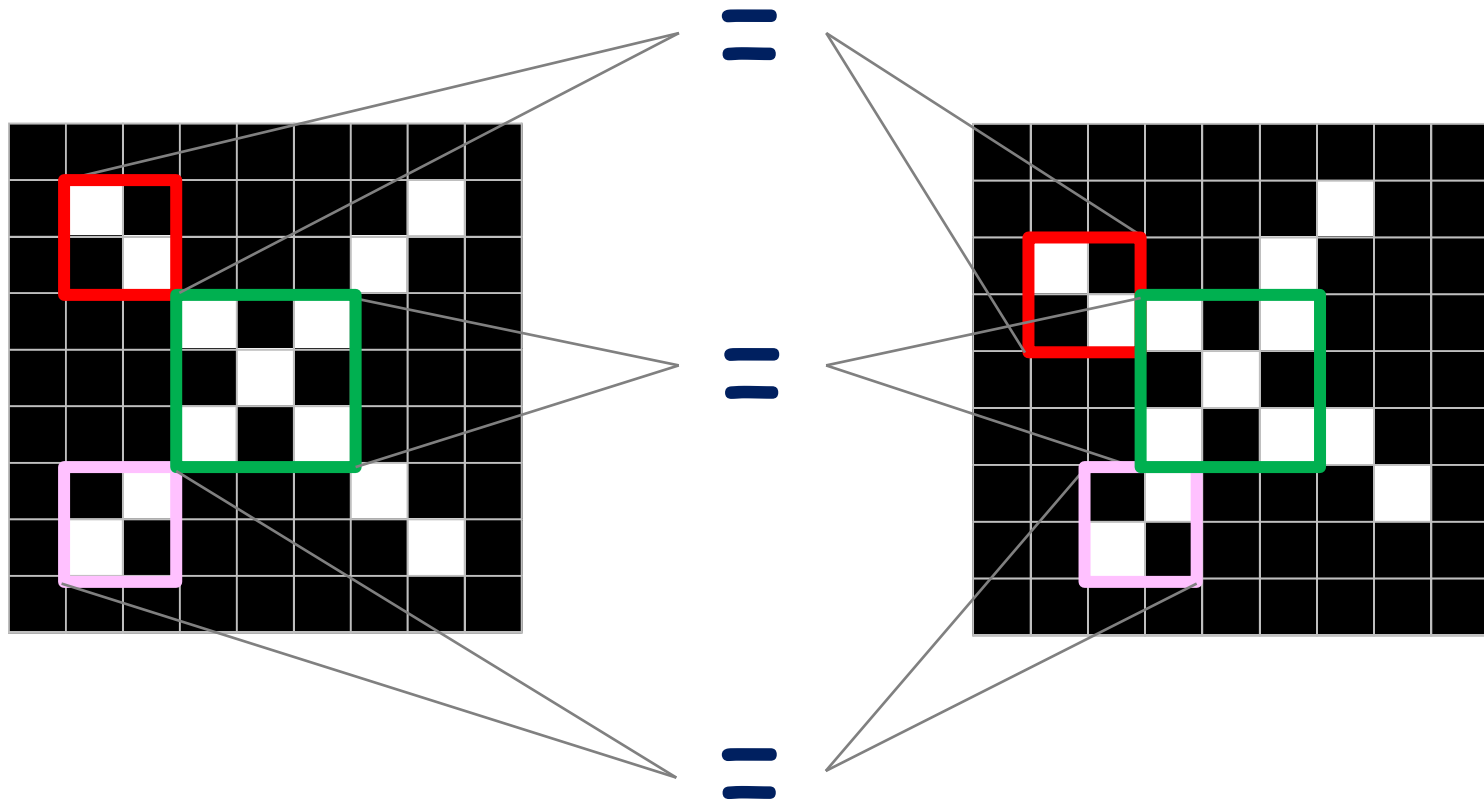


Image Classification

- **Both are partially matching**
 - Critical LOCAL features are the same



Feature Extraction

- **Convolution**

- A way to find out local features

1	0	1
0	1	1
1	1	0

 *

1	0	1
0	1	0
1	0	1

 = 4

$$I * K = \sum K_{ij} \times I_{ij}$$

Feature Extraction

■ Convolution

-1	-1	1
-1	1	-1
1	-1	-1

$$(I * K)_{xy} = \sum_{i=1}^w \sum_{j=1}^w K_{ij} \cdot I_{x+i-1, y+j-1}$$

ㄱ	ㄱ	ㄱ	ㄱ	ㄱ	ㄱ	ㄱ	ㄱ
ㄱ	0	ㄱ	1	1	ㄱ	ㄱ	ㄱ
ㄱ	ㄱ	1	ㄱ	1	1	ㄱ	ㄱ
ㄱ	1	ㄱ	ㄱ	ㄱ	ㄱ	1	ㄱ
ㄱ	1	ㄱ	ㄱ	ㄱ	ㄱ	1	ㄱ
ㄱ	ㄱ	1	ㄱ	1	1	ㄱ	ㄱ
ㄱ	ㄱ	ㄱ	1	1	ㄱ	ㄱ	ㄱ
ㄱ	ㄱ	ㄱ	ㄱ	ㄱ	ㄱ	ㄱ	ㄱ

	-1						

Feature Extraction

■ Convolution

-1	-1	1
-1	1	-1
1	-1	-1

$$(I * K)_{xy} = \sum_{i=1}^w \sum_{j=1}^w K_{ij} \cdot I_{x+i-1, y+j-1}$$

ज़	ज़	ज़	ज़	ज़	ज़	ज़	ज़
ज़	0	ज़	1	1	ज़	ज़	ज़
ज़	ज़	1	ज़	1	1	ज़	ज़
ज़	1	ज़	ज़	ज़	ज़	1	ज़
ज़	1	ज़	ज़	ज़	ज़	1	ज़
ज़	ज़	1	ज़	1	1	ज़	ज़
ज़	ज़	ज़	1	1	ज़	ज़	ज़
ज़	ज़	ज़	ज़	ज़	ज़	ज़	ज़

	-1						
	-2						

Feature Extraction

■ Convolution

-1	-1	1
-1	1	-1
1	-1	-1

$$(I * K)_{xy} = \sum_{i=1}^w \sum_{j=1}^w K_{ij} \cdot I_{x+i-1, y+j-1}$$

ㄱ	ㄱ	ㄱ	ㄱ	ㄱ	ㄱ	ㄱ	ㄱ
ㄱ	0	ㄱ	1	1	ㄱ	ㄱ	ㄱ
ㄱ	ㄱ	1	ㄱ	1	1	ㄱ	ㄱ
ㄱ	1	ㄱ	ㄱ	ㄱ	ㄱ	1	ㄱ
ㄱ	1	ㄱ	ㄱ	ㄱ	ㄱ	1	ㄱ
ㄱ	ㄱ	1	ㄱ	1	1	ㄱ	ㄱ
ㄱ	ㄱ	ㄱ	1	1	ㄱ	ㄱ	ㄱ
ㄱ	ㄱ	ㄱ	ㄱ	ㄱ	ㄱ	ㄱ	ㄱ

	-1						
	-2						
	1						

Feature Extraction

■ Convolution

-1	-1	1
-1	1	-1
1	-1	-1

$$(I * K)_{xy} = \sum_{i=1}^w \sum_{j=1}^w K_{ij} \cdot I_{x+i-1, y+j-1}$$

ㄱ	ㄱ	ㄱ	ㄱ	ㄱ	ㄱ	ㄱ	ㄱ
ㄱ	0	ㄱ	1	1	ㄱ	ㄱ	ㄱ
ㄱ	ㄱ	1	ㄱ	1	1	ㄱ	ㄱ
ㄱ	1	ㄱ	ㄱ	ㄱ	ㄱ	1	ㄱ
ㄱ	1	ㄱ	ㄱ	ㄱ	ㄱ	1	ㄱ
ㄱ	ㄱ	1	ㄱ	1	1	ㄱ	ㄱ
ㄱ	ㄱ	ㄱ	1	1	ㄱ	ㄱ	ㄱ
ㄱ	ㄱ	ㄱ	ㄱ	ㄱ	ㄱ	ㄱ	ㄱ

	-1	-2	0	-2	-1	1	
	-2	3	-2	-2	-2	-2	
	1	-1	-2	0	-4	-1	
	-1	-3	0	-2	0	1	
	-2	-1	-4	0	2	-2	
	1	-2	0	0	-3	-1	

Feature Extraction

Convolution

-1	-1	1
-1	1	-1
1	-1	-1

$$(I * K)_{xy} = \sum_{i=1}^w \sum_{j=1}^w K_{ij} \cdot I_{x+i-1, y+j-1}$$

0	0	0	0	0	0	0	0	0	0
0	ज़	ज़	ज़	ज़	ज़	ज़	ज़	ज़	0
0	ज़	0	ज़	1	1	ज़	ज़	ज़	0
0	ज़	ज़	1	ज़	1	1	ज़	ज़	0
0	ज़	1	ज़	ज़	ज़	ज़	1	ज़	0
0	ज़	1	ज़	ज़	ज़	ज़	1	ज़	0
0	ज़	ज़	1	ज़	1	1	ज़	ज़	0
0	ज़	ज़	ज़	1	1	ज़	ज़	ज़	0
0	ज़	ज़	ज़	ज़	ज़	ज़	ज़	ज़	0
0	0	0	0	0	0	0	0	0	0

0	0	-1	-2	0	1	0	0
0	-1	-2	0	-2	-1	1	0
-1	-2	3	-2	-2	-2	-2	1
-2	1	-1	-2	0	-4	-1	0
0	-1	-3	0	-2	0	1	-2
1	-2	-1	-4	0	2	-2	-1
0	1	-2	0	0	-3	-1	0
0	0	1	0	-2	-1	0	0

Feature Extraction

■ Threshold

-1	-1	1
-1	1	-1
1	-1	-1

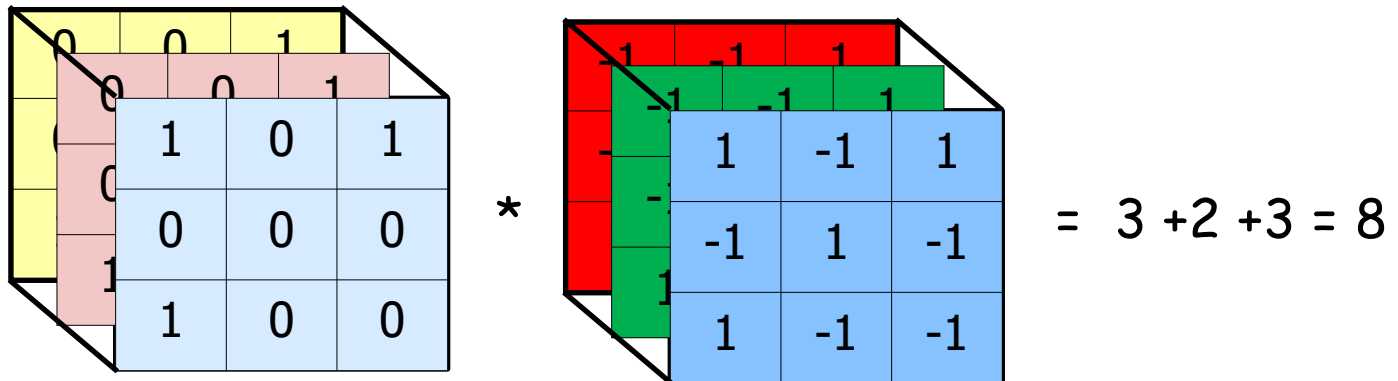
$$(I * K)_{xy} = \sum_{i=1}^w \sum_{j=1}^w K_{ij} \cdot I_{x+i-1, y+j-1}$$

ज़	ज़	ज़	ज़	ज़	ज़	ज़	ज़
ज़	0	ज़	1	1	ज़	ज़	ज़
ज़	ज़	1	ज़	1	1	ज़	ज़
ज़	1	ज़	ज़	ज़	ज़	1	ज़
ज़	1	ज़	ज़	ज़	ज़	1	ज़
ज़	ज़	1	ज़	1	1	ज़	ज़
ज़	ज़	ज़	1	1	ज़	ज़	ज़
ज़	ज़	ज़	ज़	ज़	ज़	ज़	ज़

0	0	0	0	0	1	0	0
0	0	0	0	0	0	1	0
0	0	3	0	0	0	0	1
0	1	0	0	0	0	0	0
0	0	0	0	0	0	1	0
1	0	0	0	0	2	0	0
0	1	0	0	0	0	0	0
0	0	1	0	0	0	0	0

Feature Extraction

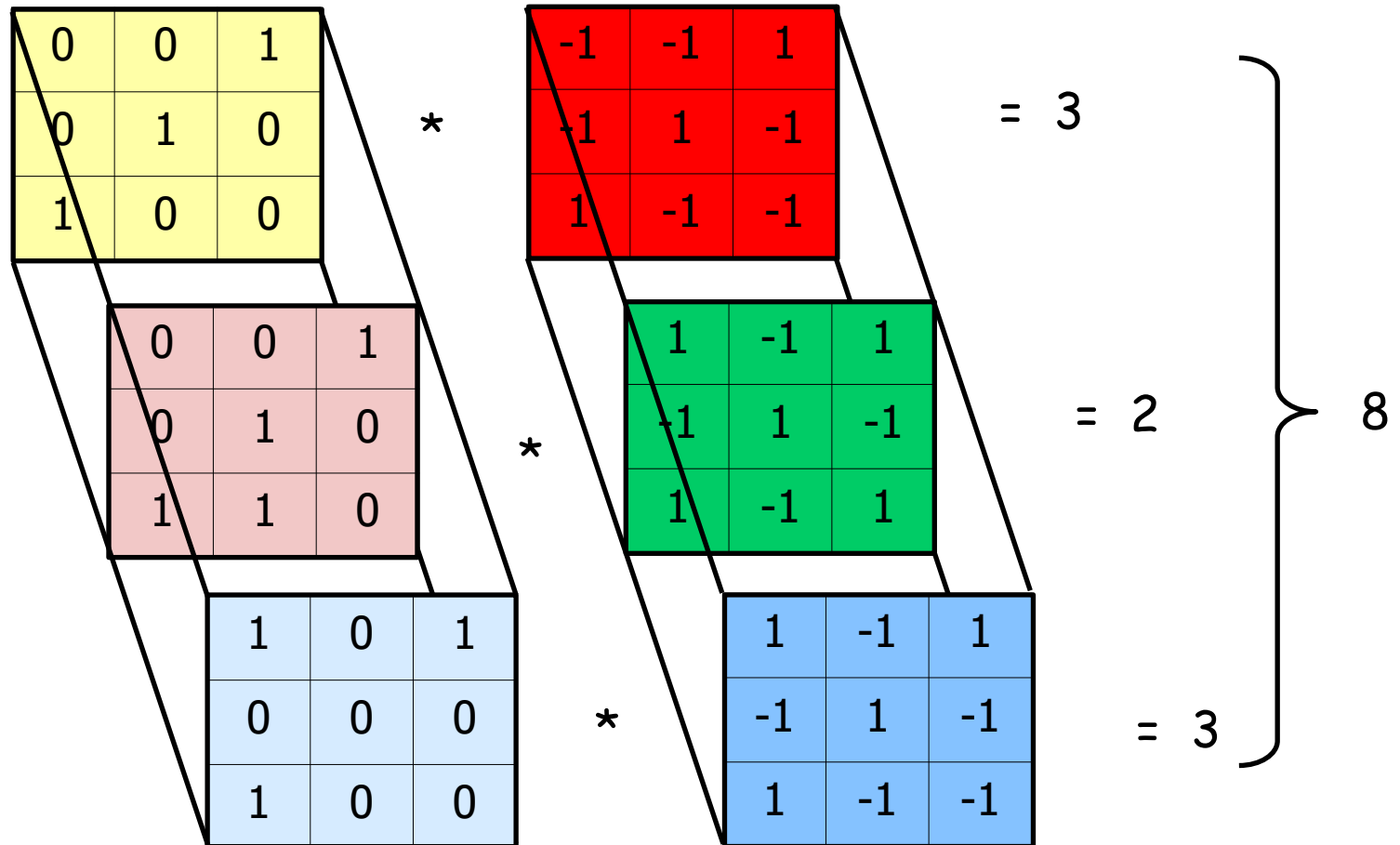
■ Convolution (3D)



$$I * K = \sum_{channel} \sum_{i,j} K_{ij}^{channel} \times I_{ij}^{channel}$$

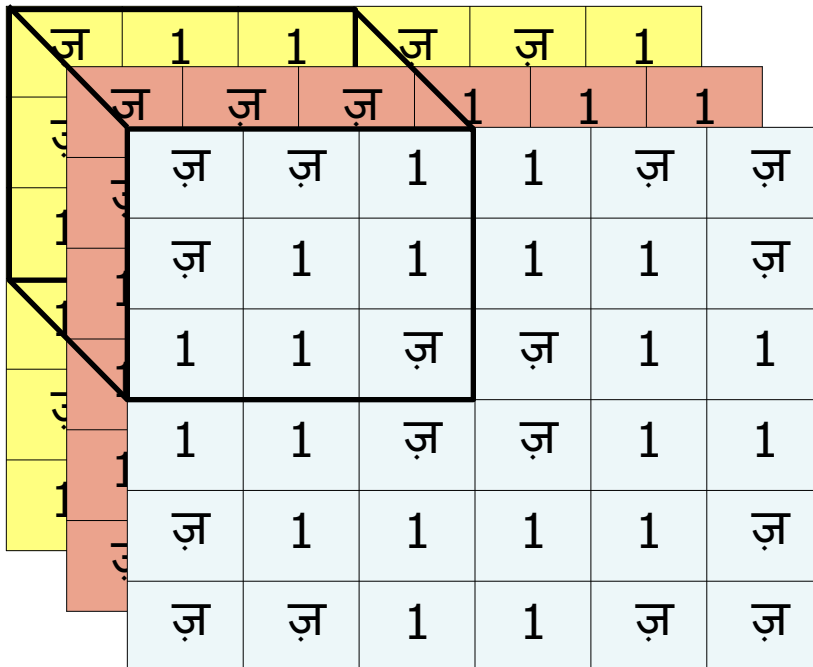
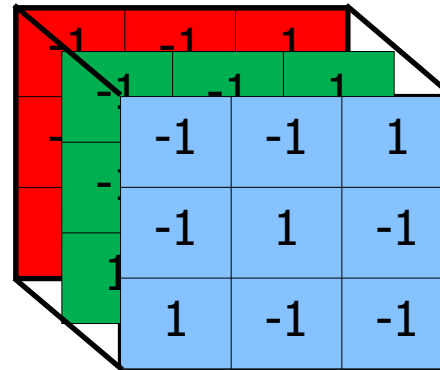
Feature Extraction

■ Convolution (3D)



Feature Extraction

Convolution (3D)



-1	2	4	-2	-1	-4
0	2	5	1	3	2
-2	0	4	2	-6	1
-2	1	0	-2	3	-3
1	2	-4	4	2	-2
1	-2	-4	3	2	1

Feature Extraction

- Examples of Convolution

1	1	1
0	0	0
-1	-1	-1

Horizontal Line



Input



Output
(Feature Map)

Feature Extraction

■ Pooling

- Subsampling from m by m pixels into 1 pixels
- Max, averaging or L^p pooling

0	0	0	0	0	1	0	0
0	0	0	0	0	0	1	0
0	0	3	0	0	0	0	1
0	1	0	0	0	0	0	0
0	0	0	0	0	0	1	0
1	0	0	0	0	2	0	0
0	1	0	0	0	0	0	0
0	0	1	0	0	0	0	0

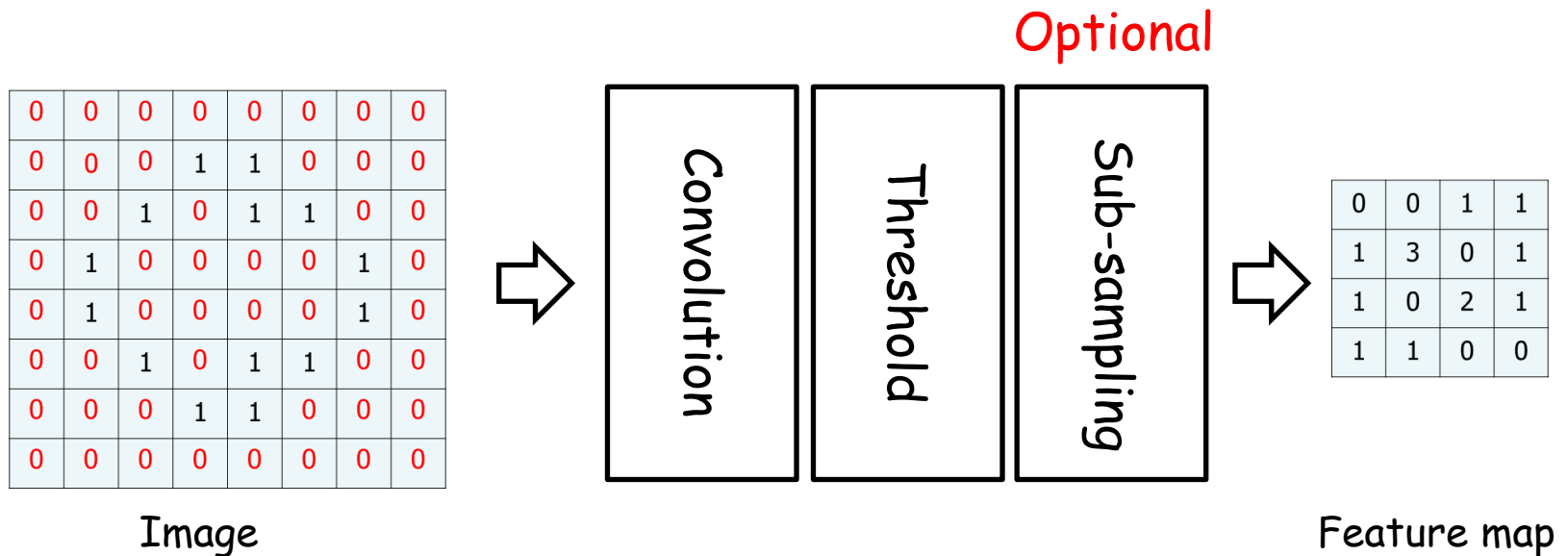
Feature map

0	0	1	1
1	3	0	1
1	0	2	1
1	1	0	0

Subsampled
feature map

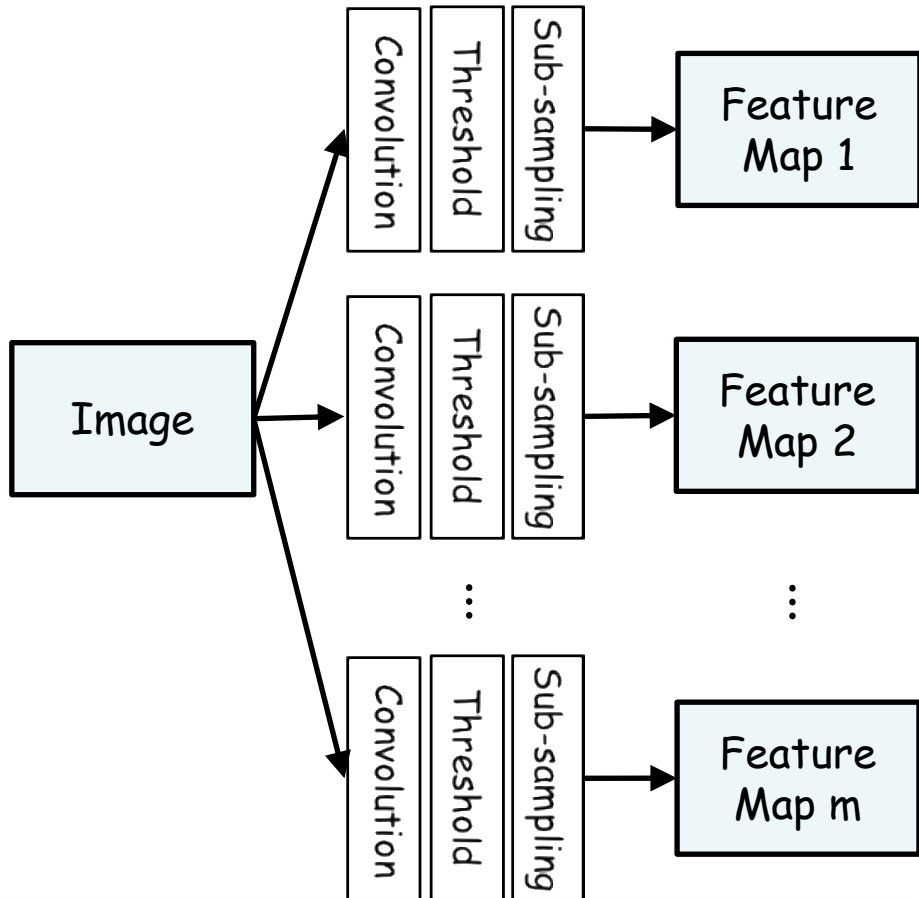
Convolutional Neural Networks

- Feature Extraction



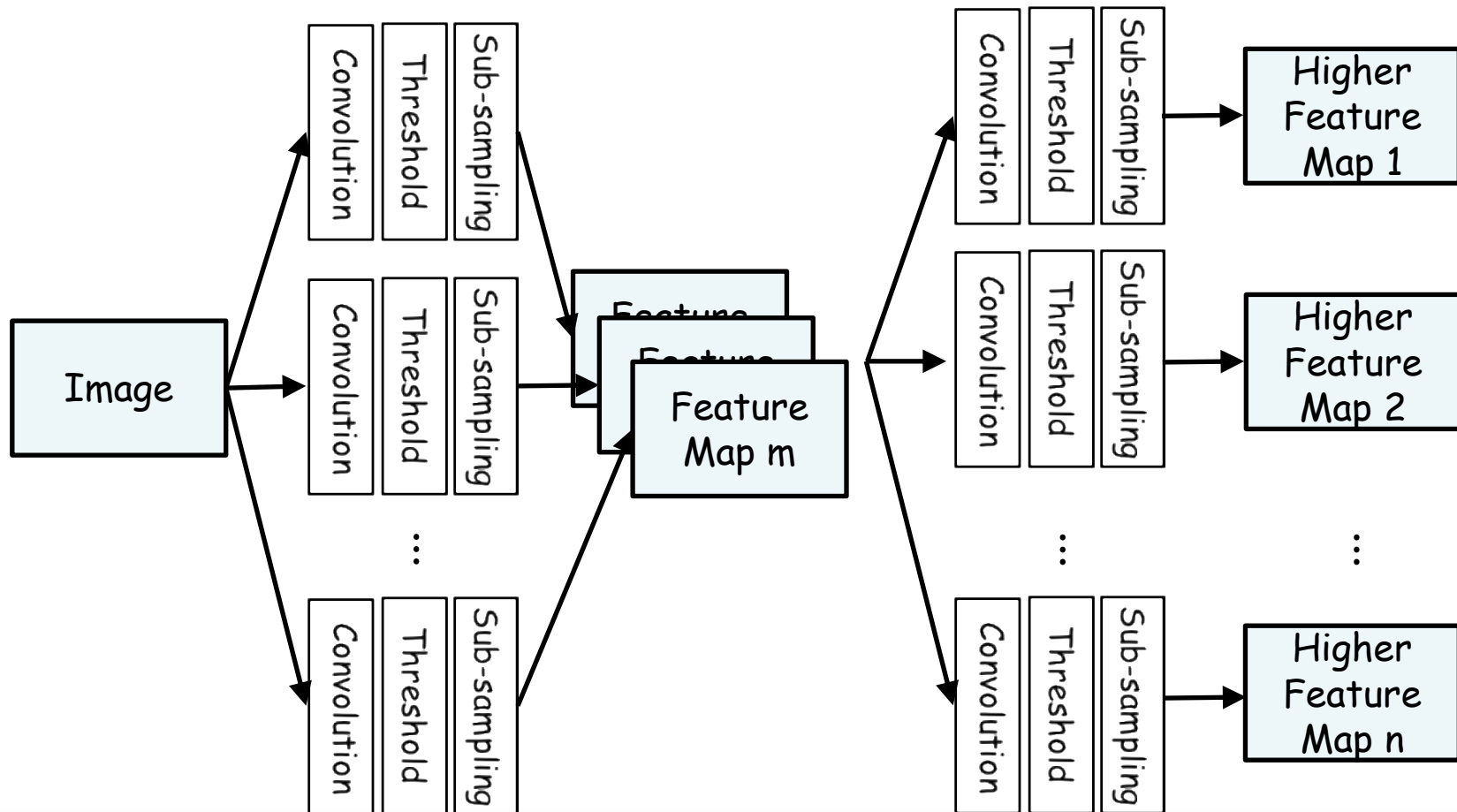
Convolutional Neural Networks

- Structure



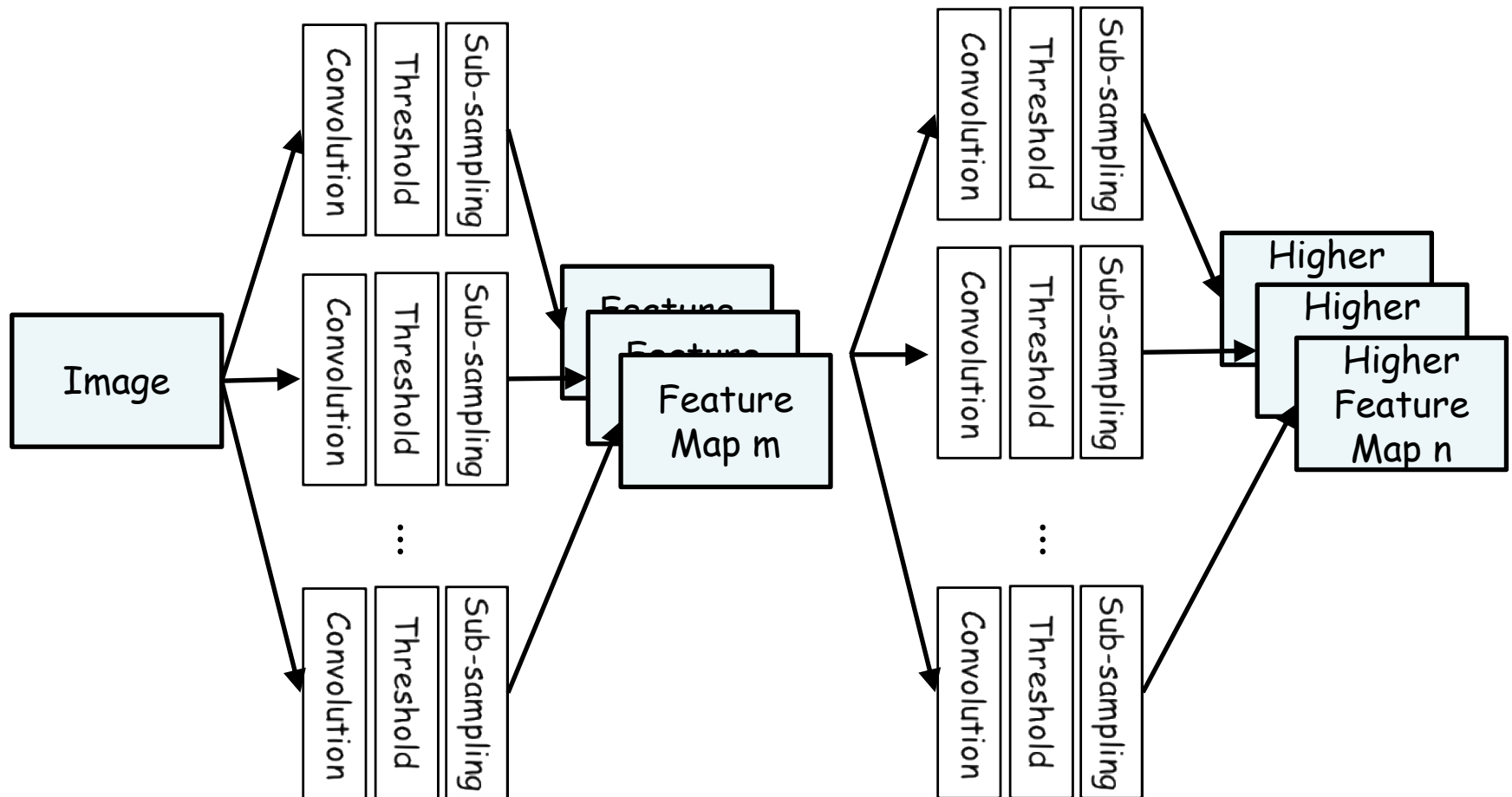
Convolutional Neural Networks

■ Structure



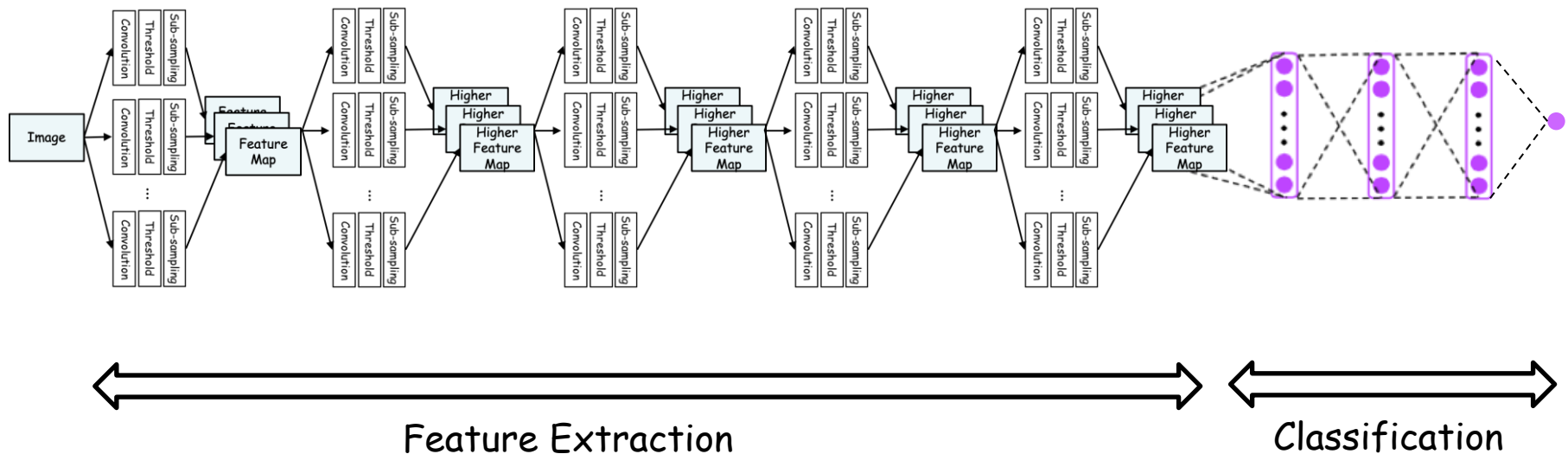
Convolutional Neural Networks

■ Structure



Convolutional Neural Networks

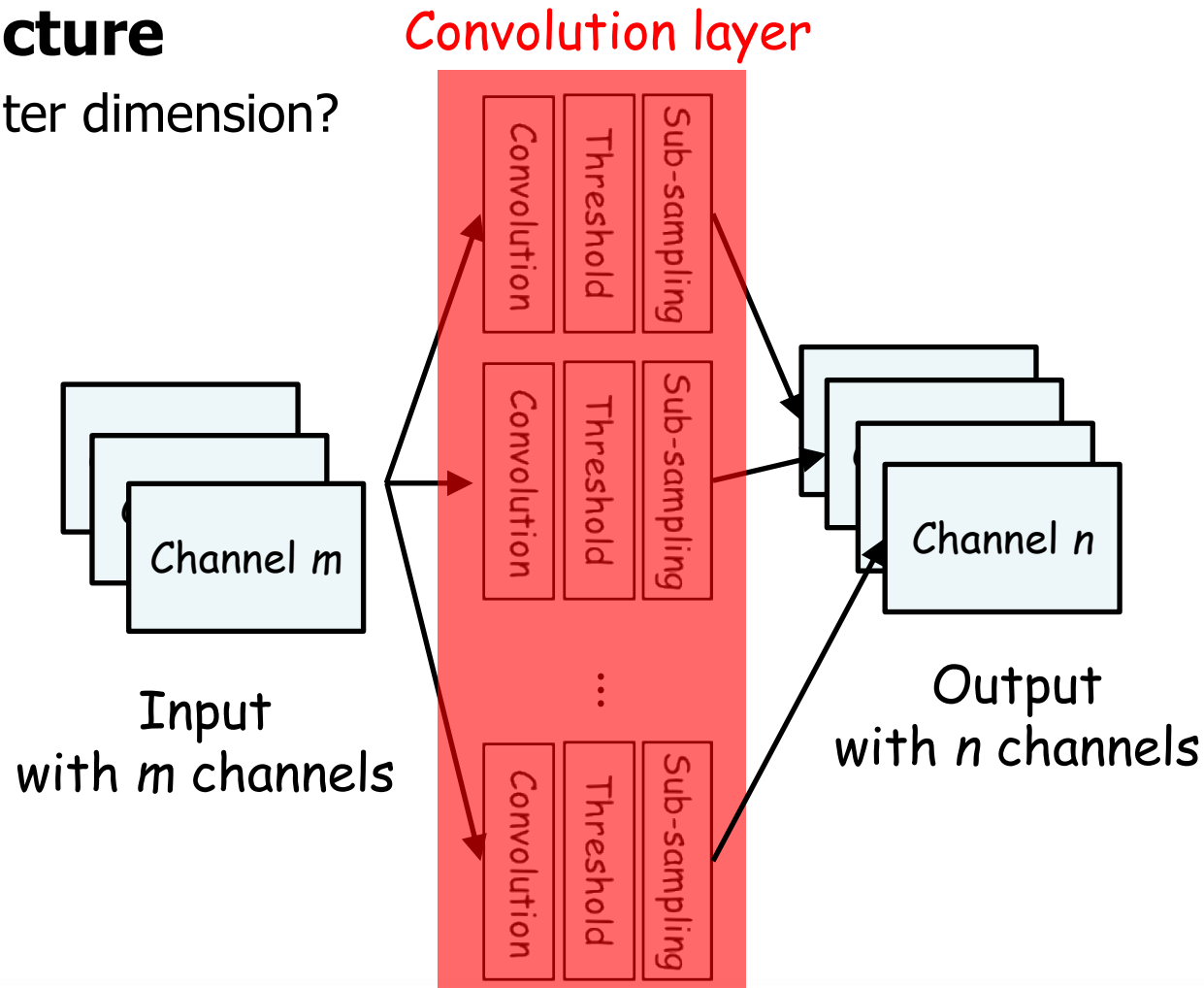
■ Structure



Convolutional Neural Networks

- **Structure**

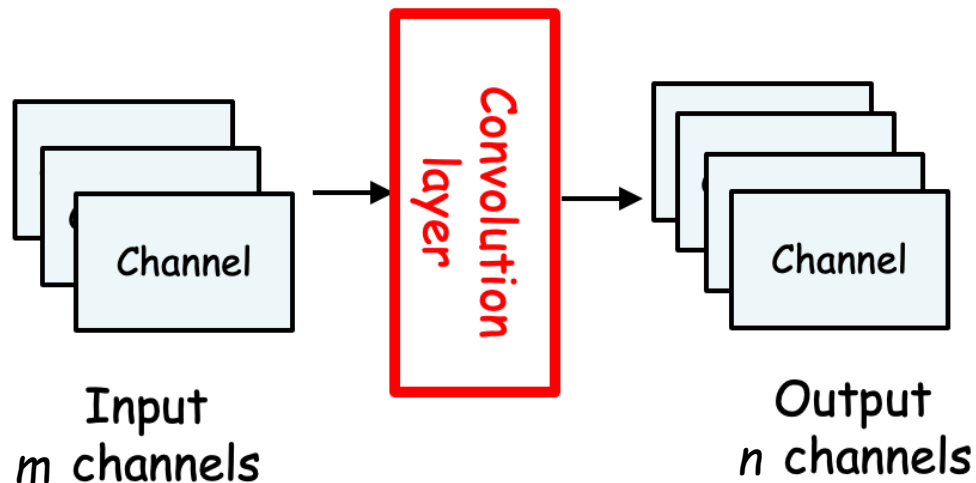
- Filter dimension?



Convolutional Neural Networks

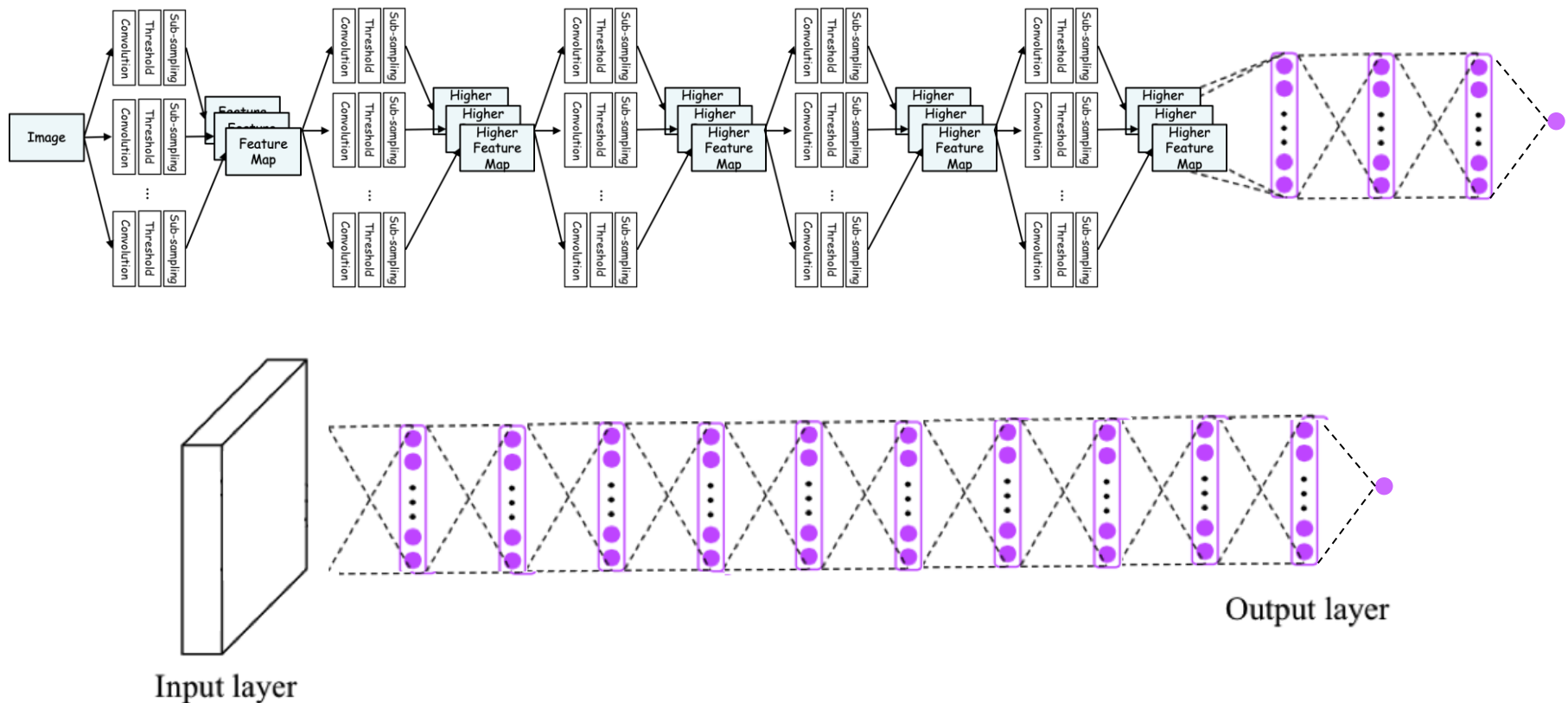
■ Structure

- How many convolutions in the convolution layer?
- What are the dimensions of each filter?
- How many trainable parameters in the conv. layer?



Convolutional Neural Networks

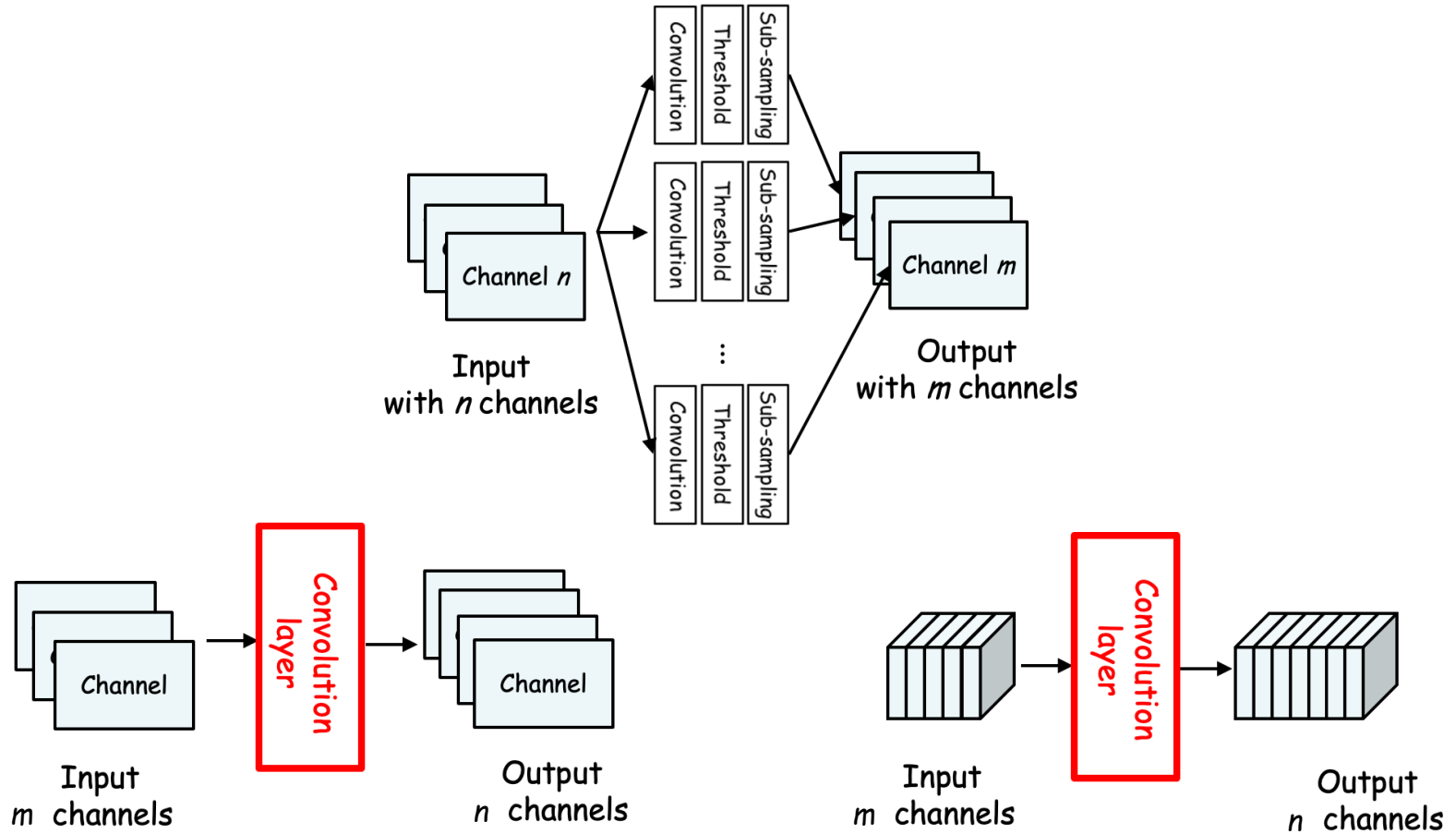
■ Structure



Convolutional Neural Networks

- **Who determines convolution masks?**
 - designed by EXPERTS!!
- **In CNN**
 - CNNs can be converted into neural networks
 - Convolution masks are converted into connection weights
 - Filters are found with gradient descent methods

Graphical Notation



Convolutional Neural Networks

- Structure

